

Scholaro's GCSE Biology Guide: Topic 3 - Infection and Response (Made for ESL Students)

Welcome to Your Infection and Response ESL Guide!

Purpose of This Guide

This guide helps ESL students understand **Topic 3 - Infection and Response** in GCSE AQA Biology. It covers diseases, how your body defends itself, how medicines work, and how we test new drugs.

What You Will Learn

-  Types of pathogens (viruses, bacteria, fungi, protists)
 -  How diseases spread and how to stop them
 -  Vaccines and antibiotics
 -  Your immune system
 -  Plant diseases and defences
 -  Drug development and monoclonal antibodies
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1. Communicable (Infectious) Diseases

Pathogens are tiny microorganisms that cause disease.

Pathogen	How it works
Virus	Enters cells and multiplies, then bursts cells
Bacteria	Divides quickly and makes toxins
Protist	Some are parasites that live inside you
Fungi	Has threads (hyphae), spreads by spores

How Diseases Spread

-  **Direct contact:** touching infected people or surfaces
-  **Water:** drinking dirty water
-  **Air:** droplets from coughs/sneezes ("droplet infection")

How to Stop the Spread

- Wash hands, clean surfaces
 - Isolate sick people
 - Kill vectors (insects)
 - Use **vaccines** to protect people
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2. Examples of Diseases

Viral Diseases

Disease	Symptoms	Spread	Prevention
Measles	Fever, red rash	Droplets in air	Vaccination
HIV/AIDS	Flu-like, then weak immune system	Blood/sex	Condoms, no needle sharing, drugs
TMV (plants)	Yellow leaves	Touch, insects	Clean fields, pest control

Bacterial Diseases

Disease	Symptoms	Spread	Prevention
Salmonella	Fever, cramps, diarrhoea	Raw food	Vaccinate poultry, cook well
Gonorrhoea	Yellow/green discharge	Sex	Use condoms, antibiotics

Fungal Disease

Disease	Symptoms	Spread	Prevention
Rose black spot	Spots on leaves	Rain/wind	Burn leaves, fungicide

Protist Disease

Disease	Symptoms	Spread	Prevention
Malaria	Fever, shaking	Mosquito bites	Insect nets, kill larvae, drugs

3. Human Defence System

Non-Specific Defence (Stops all pathogens)

Body Part	Job
Skin	Barrier, kills germs
Nose	Mucus + hairs trap germs
Trachea/Bronchi	Mucus traps, cilia waft it up
Stomach	Acid kills germs

Specific Defence (White Blood Cells)

Action	What it does
Phagocytosis	Swallows and destroys germs
Antibodies	Stick to germs, stop them, work faster next time
Antitoxins	Cancel out toxins made by bacteria

4. Vaccination

Vaccines contain **dead or weak** pathogens. Your body makes antibodies so you are protected **before** you get the real disease.

Why It Works:

- 1st time: slow, you get sick
- 2nd time: fast response, you don't get sick

Pros vs Cons:

Pros	Cons
Stops disease	Not always 100% effective

Protects community (herd immunity)

Rare side effects like fever



5. Antibiotics and Painkillers

- **Antibiotics** kill **bacteria only** inside your body
 - Do NOT work on viruses!
 - E.g. penicillin
- **Painkillers** (like aspirin) reduce **symptoms** (e.g. fever), not cause



Problem: Antibiotic Resistance

Some bacteria mutate and **survive antibiotics**. Then they multiply, and we get **superbugs**.



How to Prevent:

- Don't overuse antibiotics
 - Finish your whole course
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6. Developing New Drugs

New drugs must be:

- Safe (no bad side effects)
- Effective (they work)
- Right dose



Where Drugs Come From:

Source	Example
Plants	Aspirin (painkiller), Digitalis (heart drug)
Microbes	Penicillin (from mould)



How Drugs Are Tested:

Stage	What Happens
Preclinical	Test on cells, animals
Clinical	Test on people

- **Low dose** on healthy people (check for side effects)
 - **Higher dose** on sick people (check it works)
 - One group gets the **real drug**, one gets **placebo**
 - Can be **single blind** (only doctor knows) or **double blind** (no one knows)
 - Results checked by other scientists (**peer review**)
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7. Monoclonal Antibodies (Biology Only)

These are **identical antibodies** made in a lab. They stick to **one specific thing** (like a key to one lock).

How They're Made:

1. Get mouse white blood cells (lymphocytes)
2. Join them with tumour cells
3. Make a **hybridoma** that can divide and make antibodies
4. Collect and use the antibodies

Uses:

Use	What it Does
Pregnancy test	Shows blue line if hCG is present
Lab tests	Measures hormone or chemical in blood
Cancer treatment	Sticks to cancer cells and: • Carries drugs • Blocks growth signals • Alerts immune system

Pros vs Cons:

Pros	Cons
Targets specific cells	Can cause immune reactions
Used for tests and treatments	Expensive to make

8. Plant Diseases and Defences (Biology Only)

Signs of Plant Disease

- Yellow/discoloured leaves (TMV, magnesium deficiency)

- Spots or holes (fungus)
- Stunted growth (nitrate deficiency)
- Abnormal growths (galls)
- Pests (aphids)

How to Identify:

- Gardening books
- Lab tests
- Monoclonal antibody kits

Ion Deficiency:

Ion	Needed for	Without it...
Nitrate	Making proteins	Stunted growth
Magnesium	Making chlorophyll	Yellow leaves (chlorosis)

Plant Defence Systems

Physical:

- Waxy cuticle (blocks germs)
- Strong cell walls
- Bark or dead layers

Chemical:

- Poisons (scare animals)
- Antibacterial substances (mint, witch hazel)

Mechanical:

- Thorns and hairs
 - Leaves that curl or move
 - Mimicry (look like sick or egg-covered plants)
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Summary

You learned:

- How diseases spread and how to stop them
- How your body fights infections

- How vaccines protect you
- What antibiotics do (and don't do)
- How new medicines are made
- Plant diseases and clever defences

🎉 Great work! Keep going strong. This topic is important for understanding health, disease and how we treat them.

Made with ❤️ by Scholaro for ESL students studying AQA GCSE Biology.